lT infrastructure Management
Optimizing Pilgrim Flow: Sensor Integration and Real-time Analysis for Umrah and Hajj Management"

Group Members:

Students name	ID

Instructors:
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Introduction :

Each year, Muslims worldwide gather in Mecca, Saudi Arabia, for the sacred rites of Hajj and Umrah, integral to the Islamic faith. These rituals, observed during specific times, serve as profound journeys of spiritual devotion, fostering unity and solidarity among participants. In response to the growing number of pilgrims, the Kingdom of Saudi Arabia has undertaken significant expansions of the Two Holy Mosques and invested in crowd management initiatives to ensure the safety and comfort of visitors. Despite these efforts, challenges persist due to the sheer volume of attendees.

To address these challenges, a specialized network infrastructure has been implemented, integrating sensors and cameras to collect real-time data for analysis. This infrastructure enables optimal route planning and smooth movement within the Hajj sites, facilitating efficient crowd management.

Furthermore, the network connects relevant offices, enabling seamless information exchange and congestion management in the Grand Mosque. Continuous monitoring and analysis of crowd dynamics through strategically placed sensors and cameras further enhance crowd management effectiveness.

Through this comprehensive approach, integrating technology and infrastructure improvements, the aim is to enhance the overall Hajj experience, alleviate congestion, and promote a safe and streamlined journey for all pilgrims.

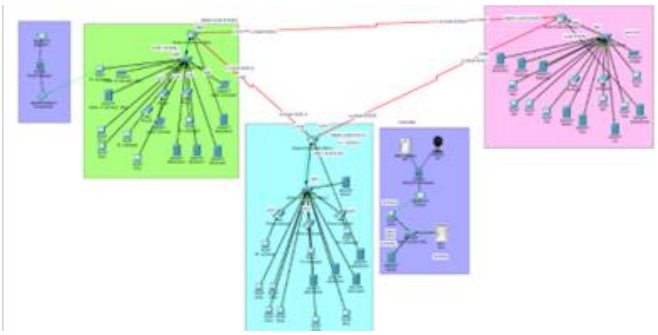
Project Goal:

Develop and set up routing protocols (RIP and OSPF) to integrate sensors and enable real-time analysis for managing Umrah and Hajj utilizing Cisco Packet Tracer, while also incorporating firewall configuration for enhancing security.

Topology of network:
Hajj pilgrimage

Network Topology:

The network for the Hajj pilgrimage uses both star and ring topologies for flexibility and reliability. Star topology allows easy scalability, while ring topology ensures communication even if a link fails. This setup ensures seamless communication for pilgrims throughout their journey.



In our network, we have implemented a star topology as the primary infrastructure to reduce costs and simplify installation. This topology allows us to connect all devices to a central hub or switch, enabling efficient management and scalability. Additionally, we have incorporated a ring topology to interconnect all the offices, providing redundancy and fault tolerance. This combination of star and ring topologies ensures a cost-effective and reliable network, facilitating seamless communication and supporting the needs of our organization.

DEVICE NAME:	QUANTITY:
Pc	More than 10
Ip phone	More than 5
routers	3
laptop	More than 2
Printer	2
server	6
switch	3
Motion detector	1
webcam	1
DLC100	2
Serial Wire Connection	3
Serial Wireless Connection	3
Straight Wire Connection	More than 20
Door 1	Door 1
Motion Detector 1	Motion Detector 1
Home Gateway 2	Home Gateway 2
Tablet 2	Tablet 2
Smart phone 1	Smart phone 1

Addressing Table:

Device	Interface	Ip address	Subnet mask	Default getaway
Router of Lost and FoundOffice	Fa0/0 S 0/0/0 S 0/0/1	10.10.30.1/24 10.10.2.1/24 10.10.3.2/24	255.255.255.0	---
Switch Of Lost And Found p2	Fa0/1-9	Take all ip address automatic from router In this pattern 10.10.30.1/24	255.255.255.0	10.10.30.1
Ip phone1	Vlan 1	10.10.30.3/24	255.255.255.0	10.10.30.1
PC6	Fa0/0	10.10.30.9/24	255.255.255.0	10.10.30.1
Dhcp server	Fa0	10.10.30.8/24	255.255.255.0	10.10.30.1
Ip phone 2	vlan	10.10.30.10/24	255.255.255.0	10.10.30.1
PC7	Fa0	10.10.30.5/24	255.255.255.0	10.10.30.1
Dns server	Fa0	10.10.30.22/24	255.255.255.0	10.10.30.1

		DNS:10.10.30.16		
PC8	Fa0	10.10.30.20/24	255.255.255.0	10.10.30.1
PC9	Fa0	10.10.30.11/24	255.255.255.0	10.10.30.1
Printer0	Fa0	10.10.30.23/24	255.255.255.0	10.10.30.1
Server-dhcp	Fa0	10.10.30.15/24 DNS:10.10.30.16	255.255.255.0	10.10.30.1
Server-dns	Fa0	10.10.30.16/24 DNS:10.10.30.16	255.255.255.0	10.10.30.1
Server-web	Fa0	10.10.30.17/24 DNS:10.10.30.16	255.255.255.0	10.10.30.1
PC0	Fa0	10.10.30.12/24	255.255.255.0	10.10.30.1
PC4	Fa0	10.10.30.19/24	255.255.255.0	10.10.30.1
PC3	Fa0	10.10.30.18/24	255.255.255.0	10.10.30.1
Server3	Fa0	10.10.30.13/24	255.255.255.0	10.10.30.1
Server-Email2	Fa0	10.10.31.2/24	255.255.255.0	10.10.31.1
Router Of Information Office_	Fa0 /0 S 0/1/0 S 0/1/1	10.10.20.1 10.10.1.2 10.10.3.1	255.255.255.0	----
Switch Of Information Office	Fa0/1-14	Take all ip address automatic from router In this pattern 10.10.20.1/24	255.255.255.0	--
PC_employee3	Fa0	10.10.20.7/24	255.255.255.0	10.10.20.1
Printer_of_information_office	Fa0	10.10.20.11/24	255.255.255.0	10.10.20.1
Server_of_information_office	Fa0	10.10.20.9/24	255.255.255.0	10.10.20.1

PC13	Fa0	10.10.20.12/24	255.255.255.0	10.10.20.1
PC11	Fa0	10.10.20.13/24	255.255.255.0	10.10.20.1
phone3	Fa0	10.10.20.6/24	255.255.255.0	10.10.20.1
PC_employee2	Fa0	10.10.20.3/24	255.255.255.0	10.10.20.1
Phone2	Fa0	10.10.20.8/24	255.255.255.0	10.10.20.1
Laptop_employee	Fa0	10.10.20.10/24	255.255.255.0	10.10.20.1
Pc10	Fa0	10.10.20.14/24	255.255.255.0	10.10.20.1
Server-dhcp1	Fa0	10.10.20.15/24	255.255.255.0	10.10.20.1
Phone1	Fa0	10.10.20.5/24	255.255.255.0	10.10.20.1
PC_employee1	Fa0	10.10.20.4/24	255.255.255.0	10.10.20.1
Server-dns1	Fa0	10.10.20.16/24	255.255.255.0	10.10.20.1
Server-web1	Fa0	10.10.20.17/24	255.255.255.0	10.10.20.1
Server-Email	Fa0	10.10.22.2/24	255.255.255.0	10.10.22.1
Laptop_employee2	Fa0	10.10.20.2/24	255.255.255.0	10.10.20.1
Router Of Information Office 2	Fa0 /0 S 0/1/0 S 0/1/1	10.10.10.1/24 10.10.1.1/24 10.10.2.2/24	---	-----
Switch Of Information Office 2	Fa0-1-14	Take all ip address automatic from router In this pattern 10.10.20.1/24	255.255.255.0	10.10.10.1
Phone employee1	Fa0	10.10.10.4/24	255.255.255.0	10.10.10.1
PC employee1	Fa0	10.10.10.7/24	255.255.255.0	10.10.10.1
Pc16	Fa0	10.10.10.11/24	255.255.255.0	10.10.10.1
Pc15	Fa0	10.10.10.12/24	255.255.255.0	10.10.10.1
Pc17	Fa0	10.10.10.9/24	255.255.255.0	10.10.10.1
Pc18	Fa0	10.10.10.10/24	255.255.255.0	10.10.10.1
Phone employee2	Fa0	10.10.10.2/24	255.255.255.0	10.10.10.1
PC employee2	Fa0	10.10.10.3/24	255.255.255.0	10.10.10.1

Pc19	Fa0	10.10.10.13/24	255.255.255.0	10.10.10.1
Server-dhcp2	Fa0	10.10.10.15/24	255.255.255.0	10.10.10.1
Pc14	Fa0	10.10.10.14/24	255.255.255.0	10.10.10.1
Server-dns2	Fa0	10.10.10.16/24	255.255.255.0	10.10.10.1
Server-web2	Fa0	10.10.10.17/24	255.255.255.0	10.10.10.1
PC employee3	Fa0	10.10.10.5/24	255.255.255.0	10.10.10.1
Phone employee3	Fa0	10.10.10.6/24	255.255.255.0	10.10.10.1
Server1	Fa0	10.10.10.8/24	255.255.255.0	10.10.10.1
Server-Email1	Fa0	10.10.12.2/24	255.255.255.0	10.10.12.1
Office of control			255.255.255.0	
Server2	Fa0	10.10.12.2/24	255.255.255.0	10.10.70.1
PC12	Fa0	10.10.12.2/24	255.255.255.0	10.10.70.2
Door	Fa0	10.10.12.2/24	255.255.255.0	10.10.70.3

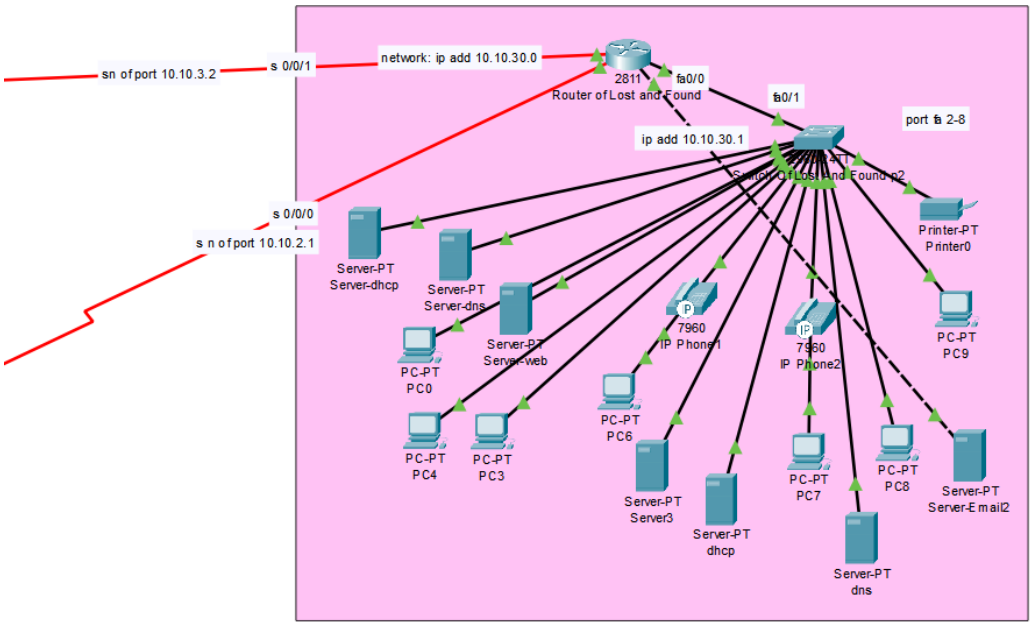
Configuration:

In Masjid al-Haram, there is a main office dedicated to lost and found services, along with 22 information offices that provide assistance to Umrah pilgrims and Hajj pilgrims.

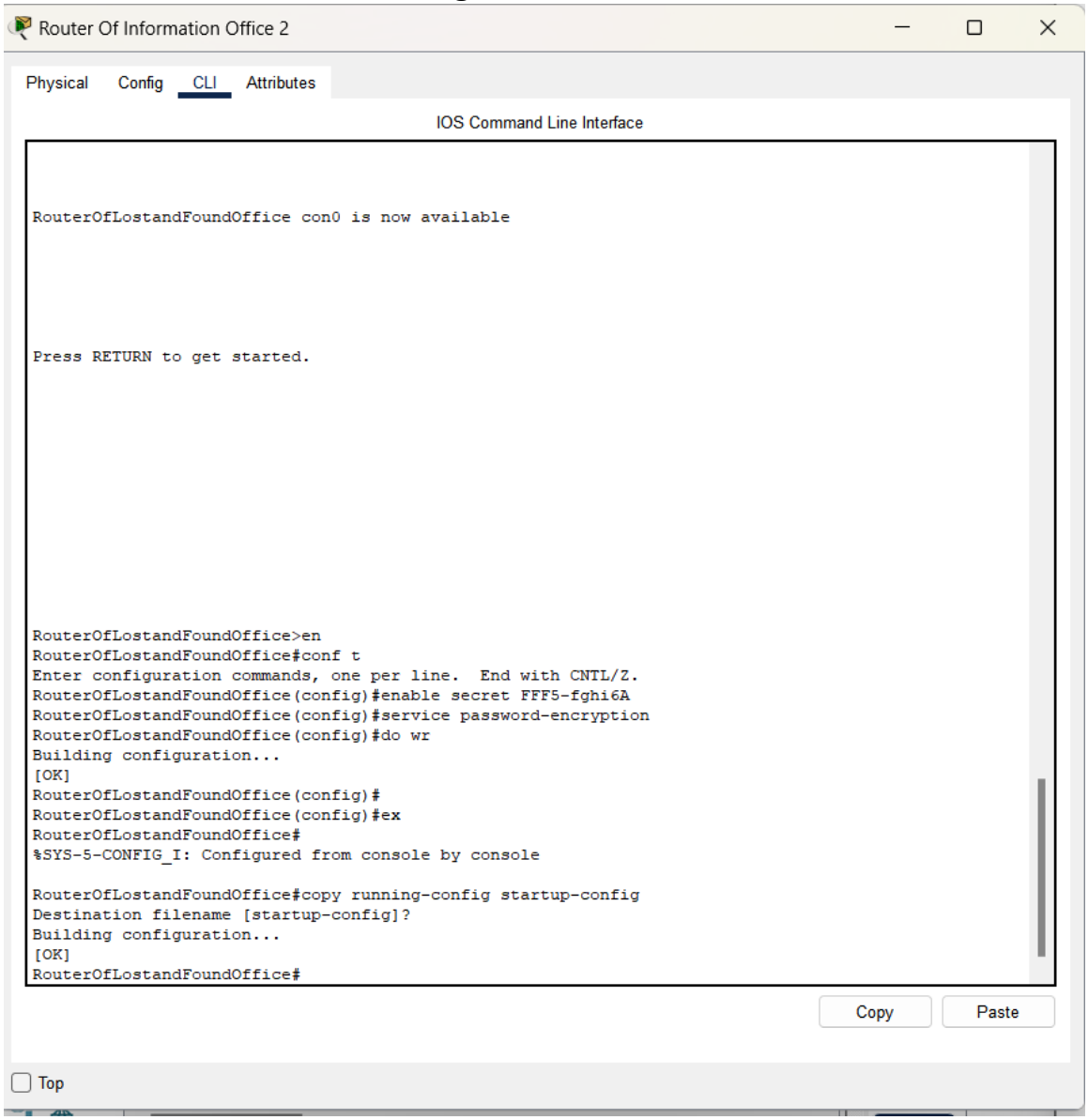
We have configured the CLI interfaces on all three routers in the main office as follows:

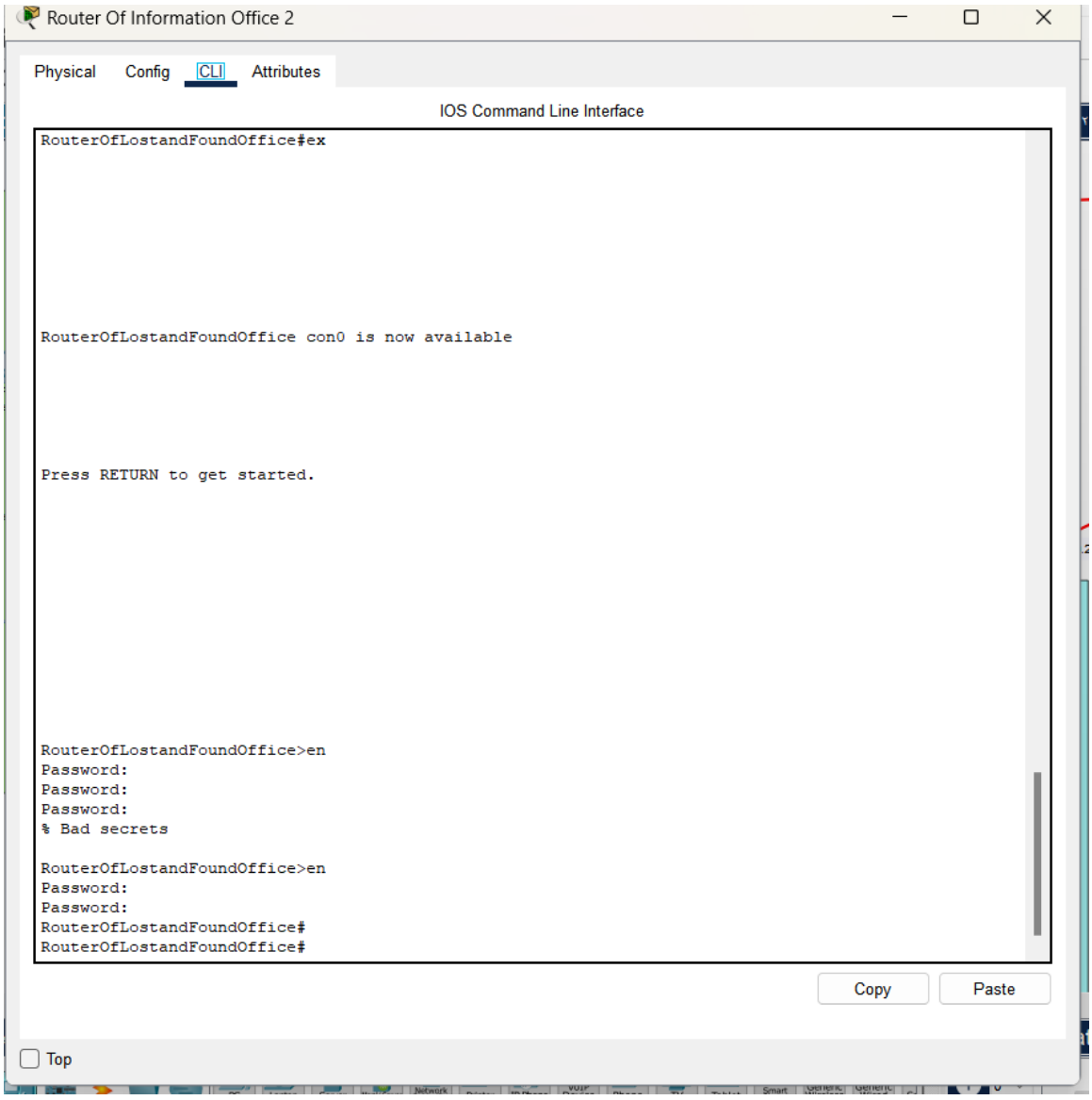
Router of Lost and Found Office (one office):

- Interface 1:Fast Ethernet connection with IP address 10.10.30.1
- Interface 2: Serial connection with IP address 10.10.3.2 and 10.10.2.1

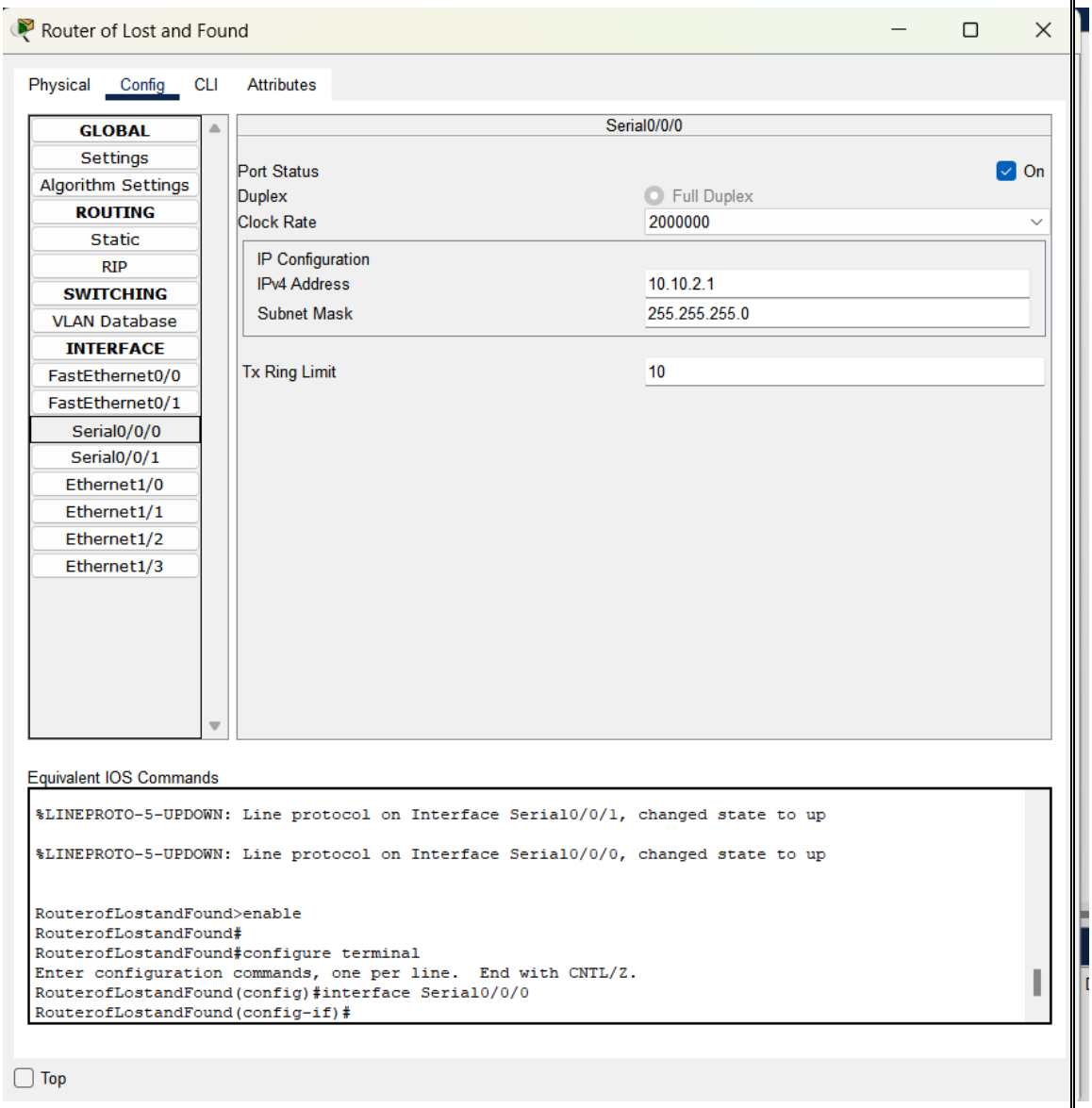


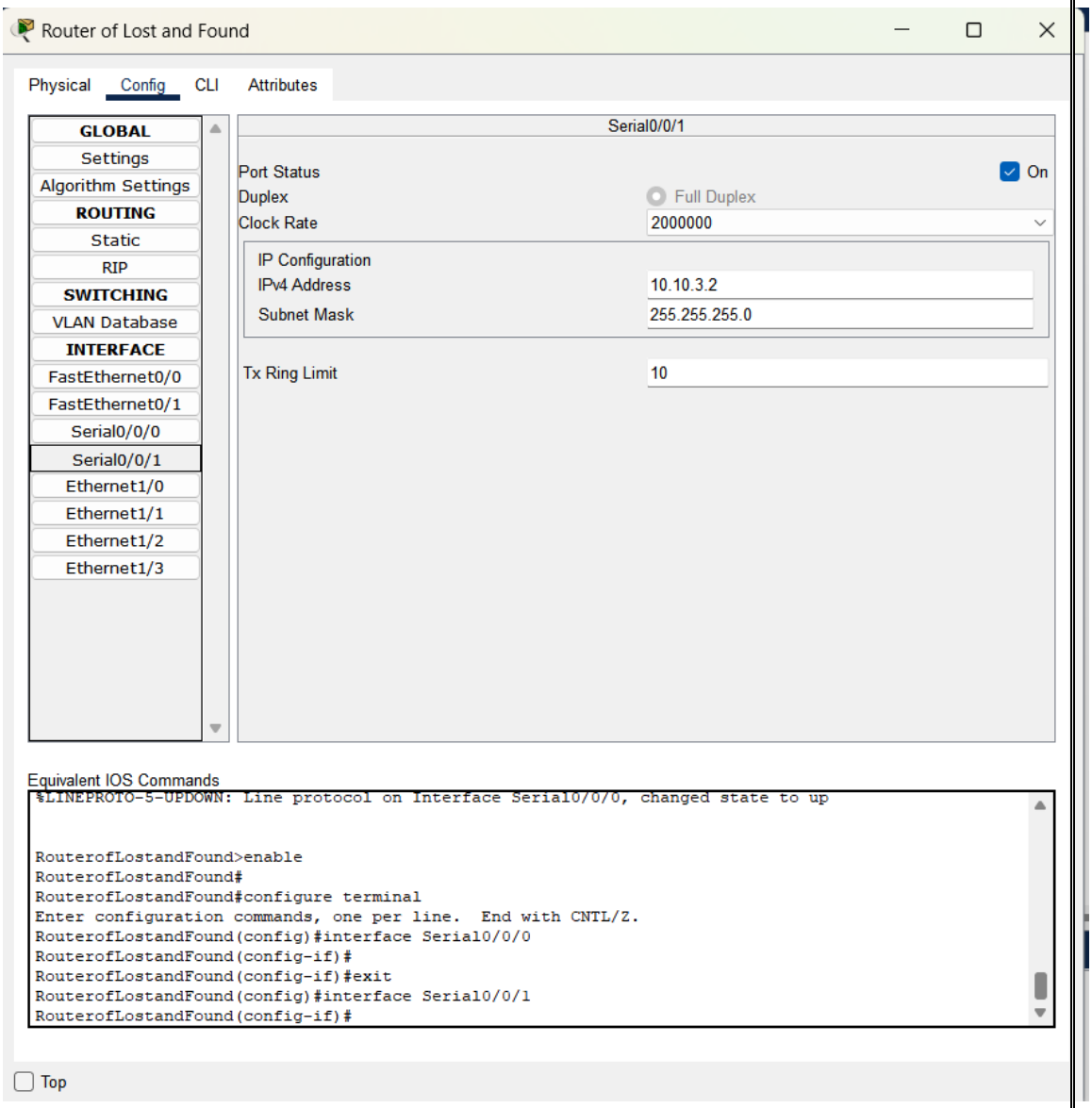
- **Router configuration:**





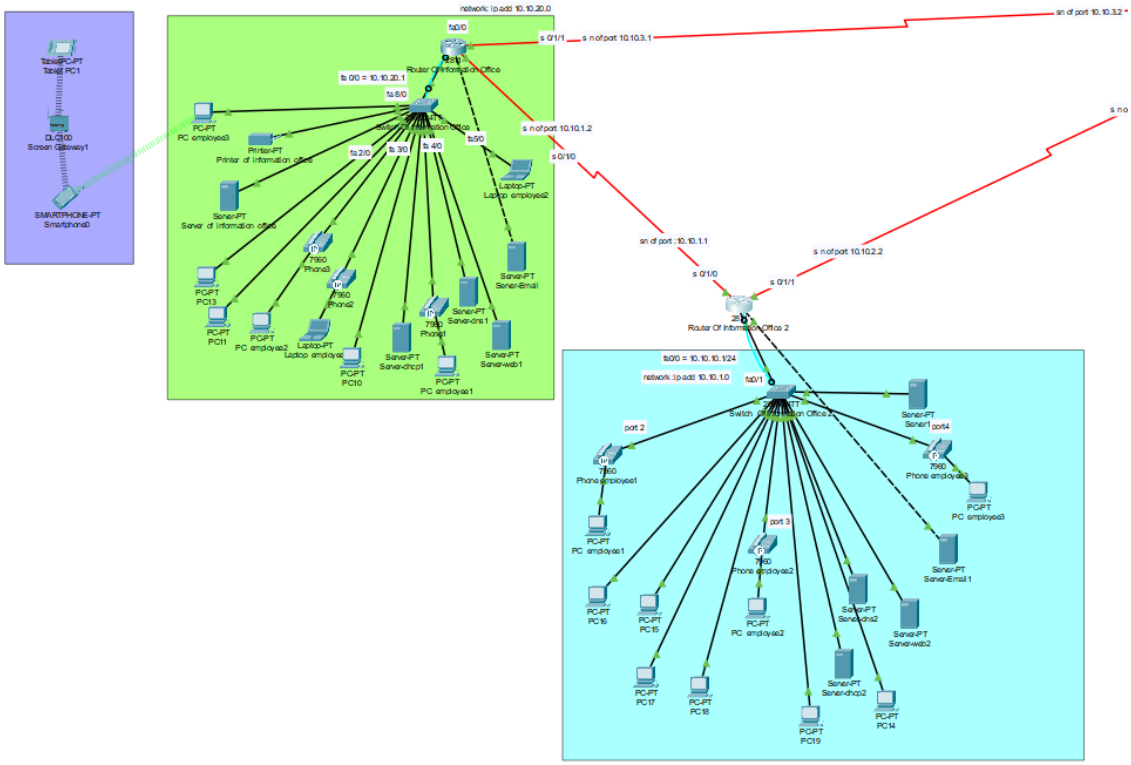
- **serial connection :**

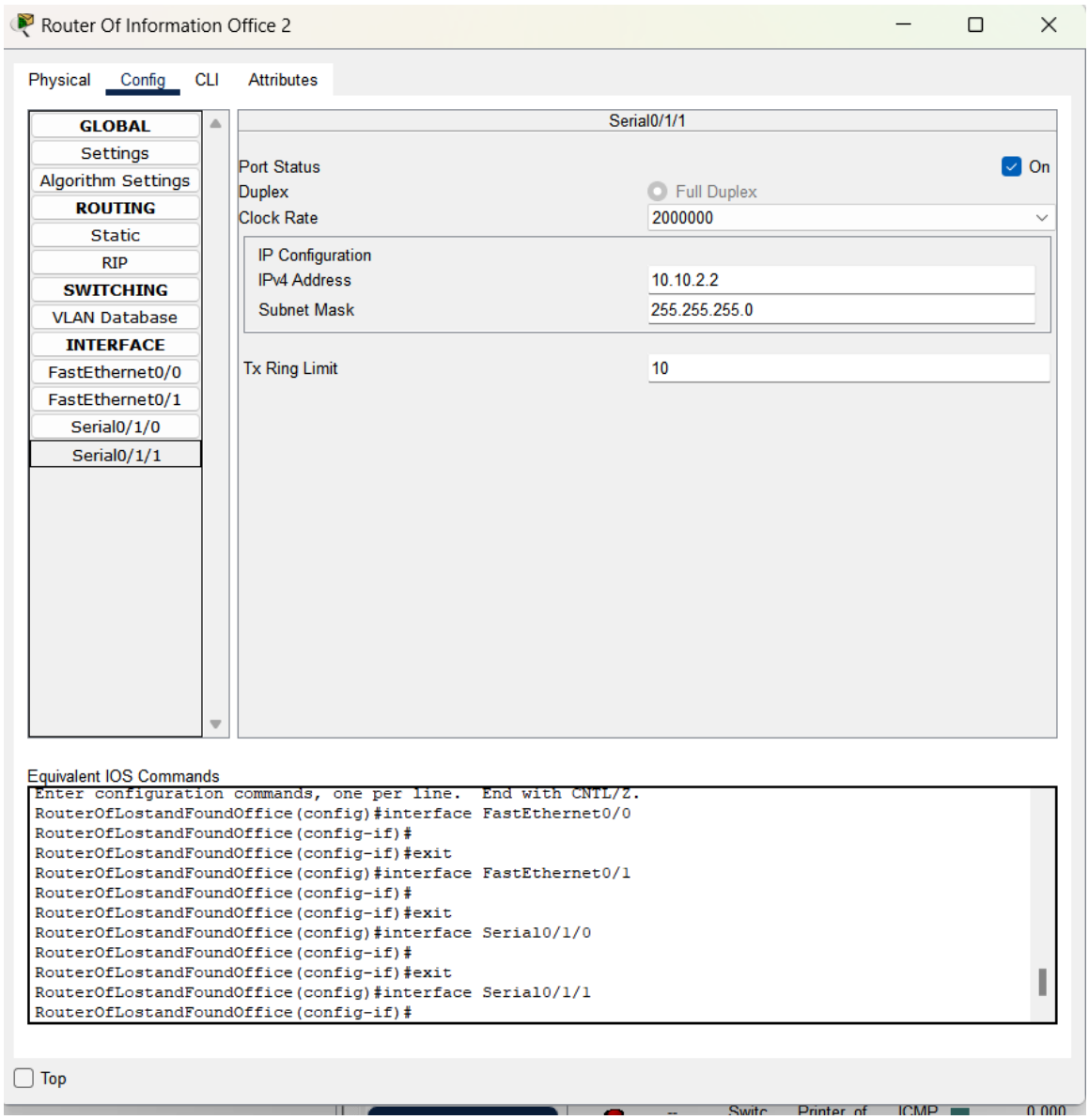

Router of Lost and Found
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Ethernet1/0
Ethernet1/1
Ethernet1/2
Ethernet1/3
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.10.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
RouterofLostandFound>enable
RouterofLostandFound#
RouterofLostandFound#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterofLostandFound(config)#interface Serial0/0/0
RouterofLostandFound(config-if)#
Top

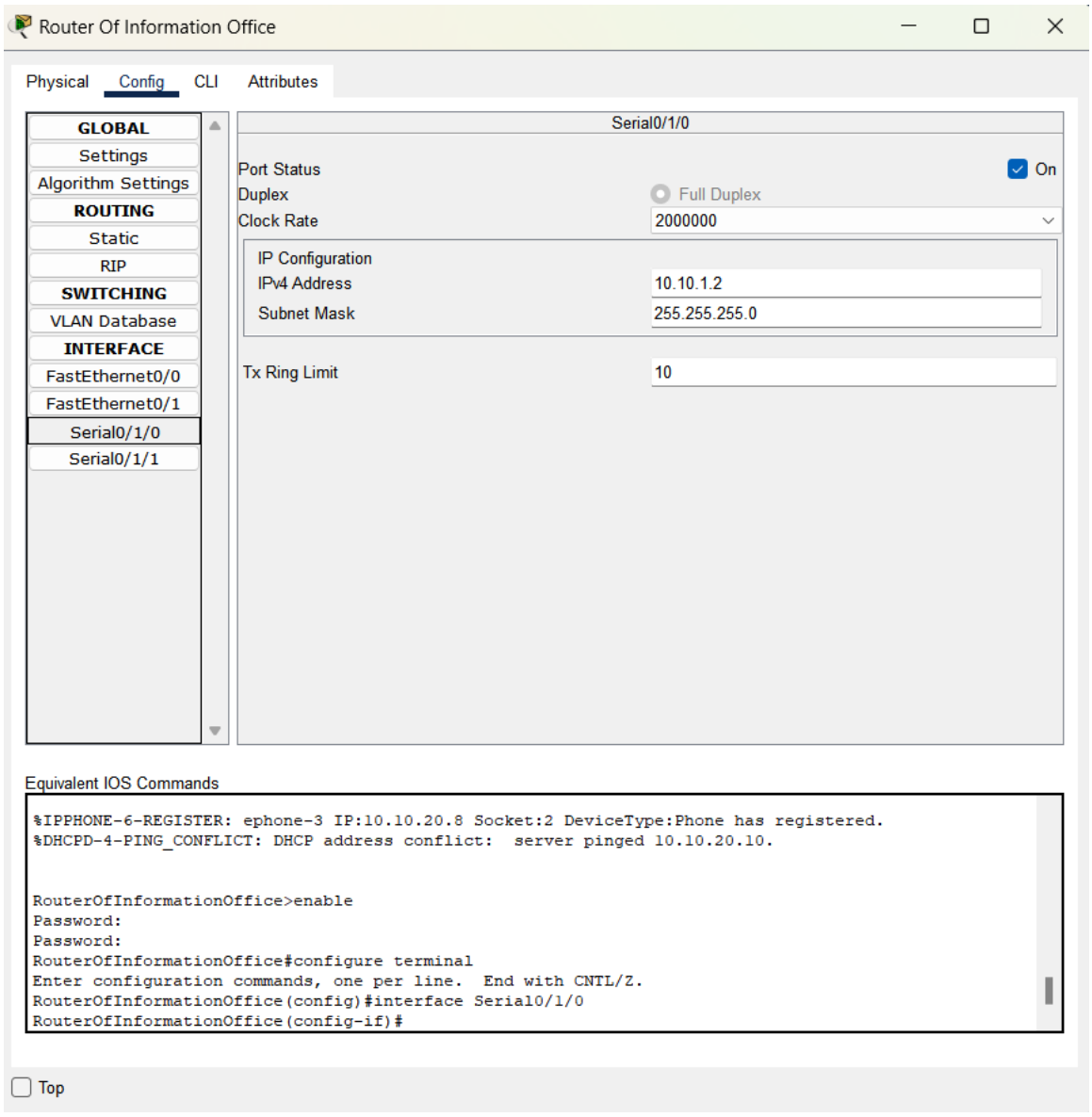

Router of Lost and Found
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Ethernet1/0
Ethernet1/1
Ethernet1/2
Ethernet1/3
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.10.3.2
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
RouterofLostandFound>enable
RouterofLostandFound#
RouterofLostandFound#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterofLostandFound(config)#interface Serial0/0/0
RouterofLostandFound(config-if)#
RouterofLostandFound(config-if)#exit
RouterofLostandFound(config)#interface Serial0/0/1
RouterofLostandFound(config-if)#
Top

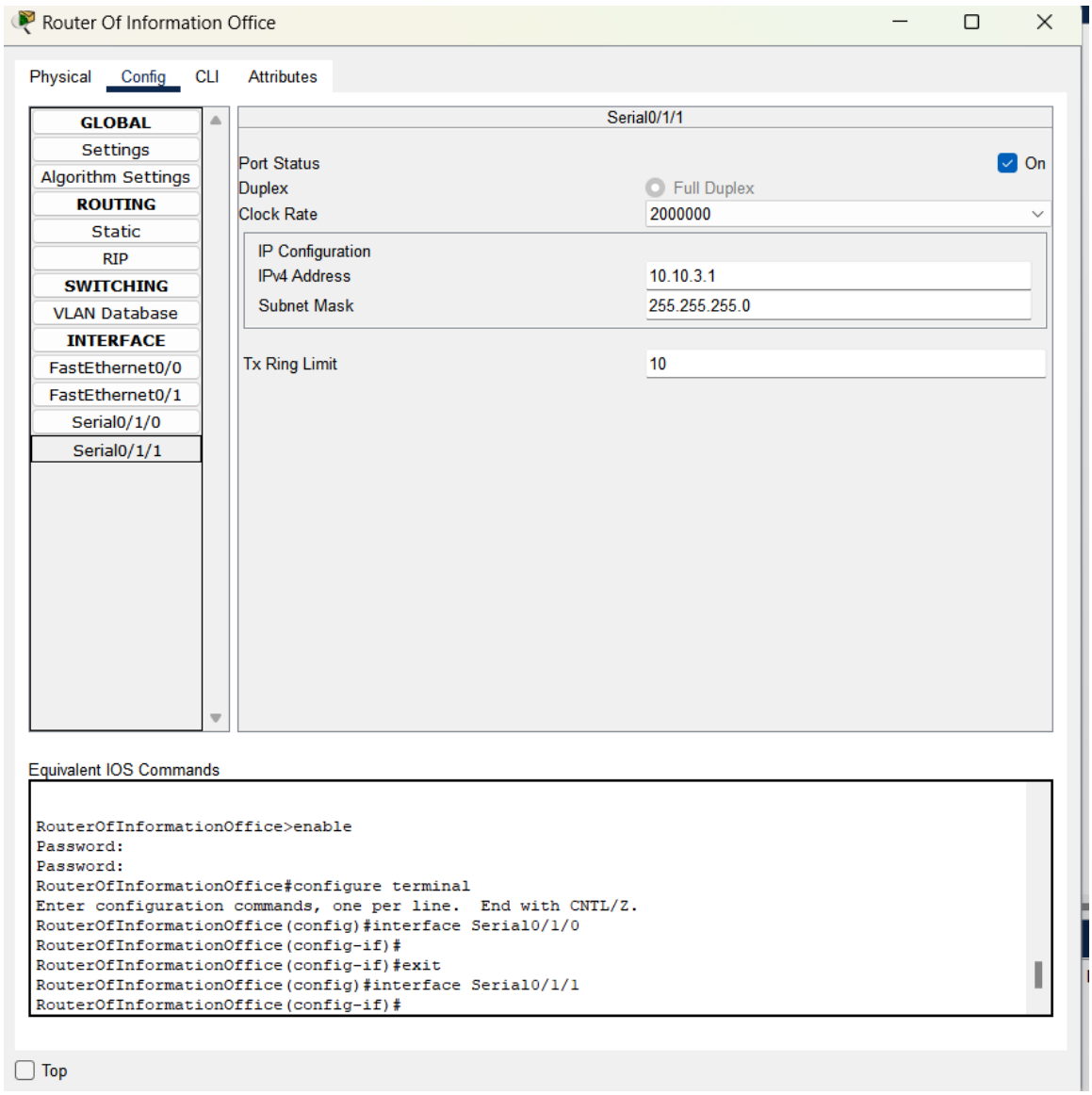
Router Of Information Office(two office):

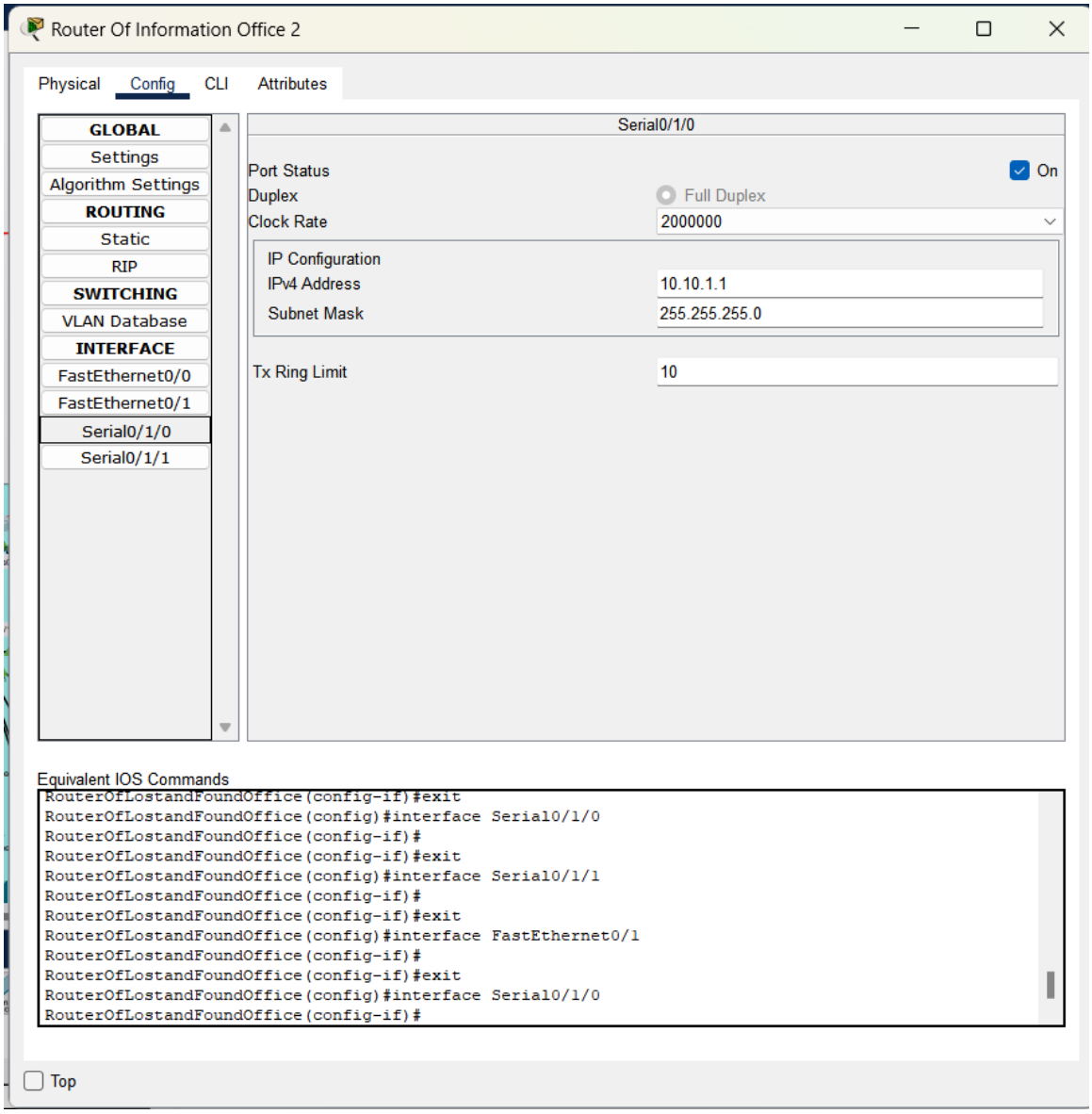
- Interface 1:Fast Ethernet connection with IP address 10.10.20.1
- Interface 2: Serial connection with IP address 10.10.1.2 and 10.10.3.1




Router Of Information Office 2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.10.2.2
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Enter configuration commands, one per line. End with CNTL/Z.
RouterOfLostandFoundOffice(config)#interface FastEthernet0/0
RouterOfLostandFoundOffice(config-if)#
RouterOfLostandFoundOffice(config-if)#exit
RouterOfLostandFoundOffice(config)#interface FastEthernet0/1
RouterOfLostandFoundOffice(config-if)#
RouterOfLostandFoundOffice(config-if)#exit
RouterOfLostandFoundOffice(config)#interface Serial0/1/0
RouterOfLostandFoundOffice(config-if)#
RouterOfLostandFoundOffice(config-if)#exit
RouterOfLostandFoundOffice(config)#interface Serial0/1/1
RouterOfLostandFoundOffice(config-if)#
Top

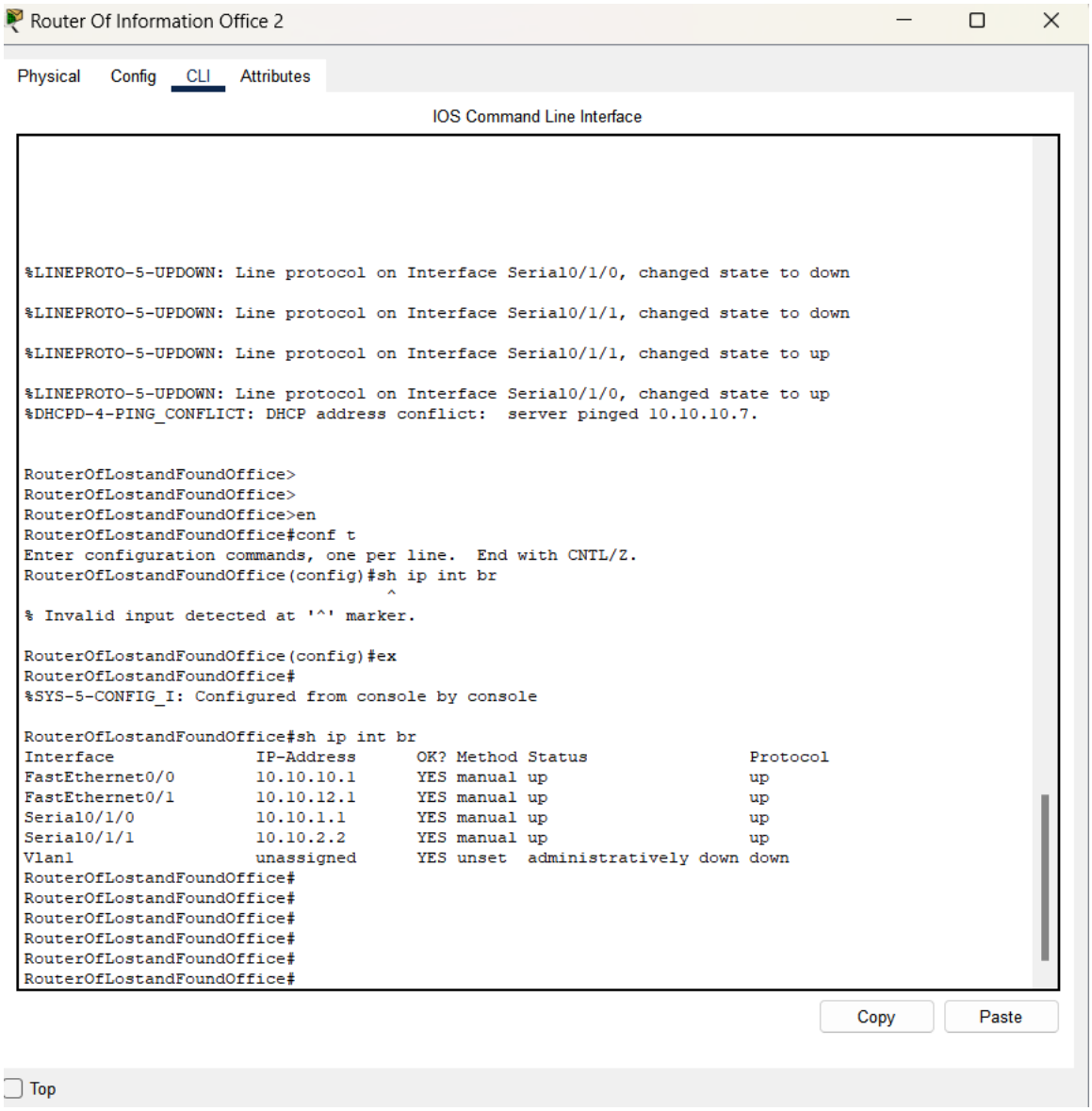

Router Of Information Office
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.10.1.2
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
%IPPHONE-6-REGISTER: ephone-3 IP:10.10.20.8 Socket:2 DeviceType:Phone has registered.
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged 10.10.20.10.
RouterOfInformationOffice>enable
Password:
Password:
RouterOfInformationOffice#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterOfInformationOffice(config)#interface Serial0/1/0
RouterOfInformationOffice(config-if)#
Top

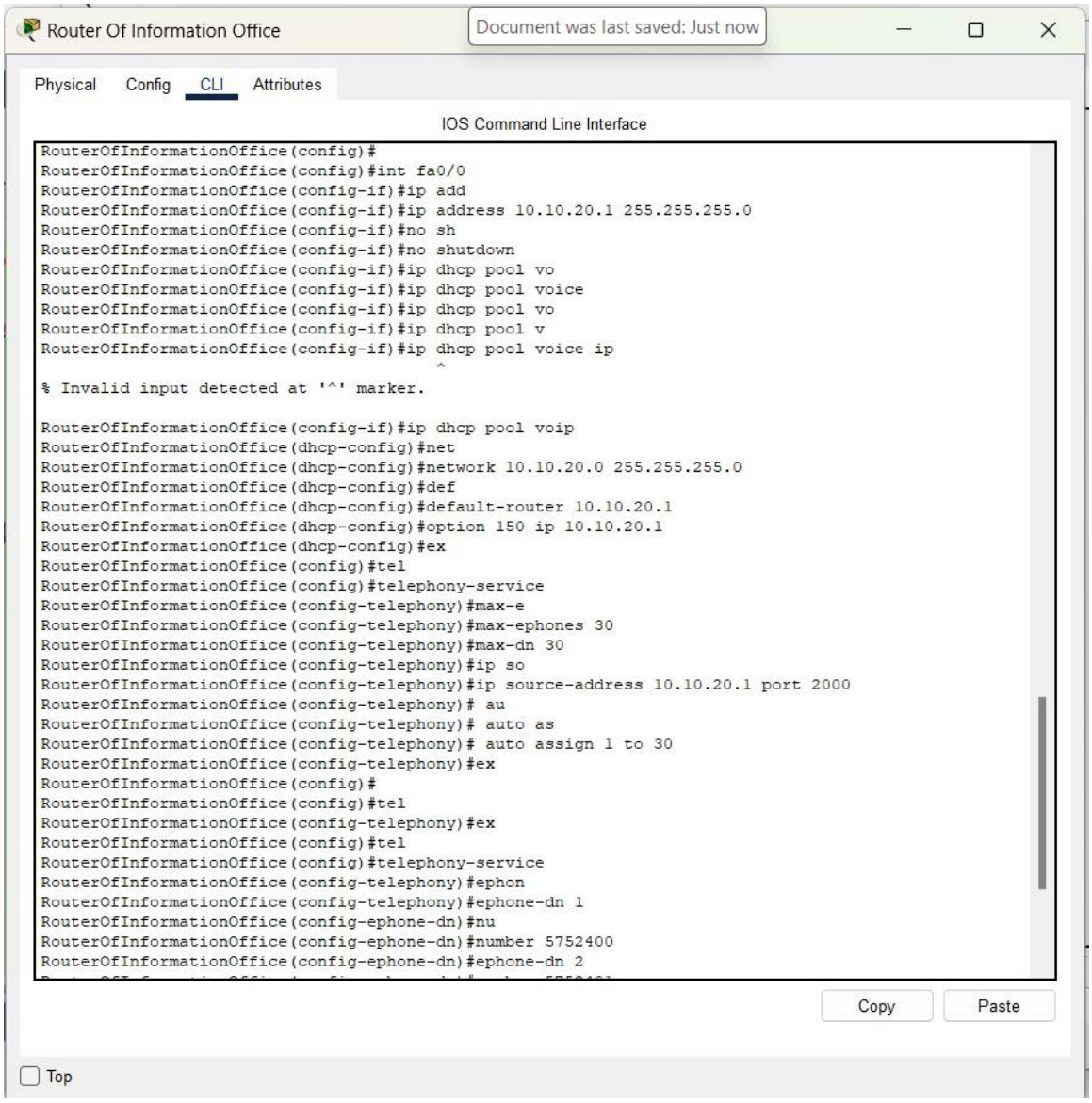

Router Of Information Office
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.10.3.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
RouterOfInformationOffice>enable
Password:
Password:
RouterOfInformationOffice#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterOfInformationOffice(config)#interface Serial0/1/0
RouterOfInformationOffice(config-if)#
RouterOfInformationOffice(config-if)#exit
RouterOfInformationOffice(config)#interface Serial0/1/1
RouterOfInformationOffice(config-if)#
Top



DHCP Protocol :

In the accounts router we used the DHCP protocol to facilitate assigning address to the devices.


Router Of Information Office 2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
%DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged 10.10.10.7.
RouterOfLostandFoundOffice>
RouterOfLostandFoundOffice>
RouterOfLostandFoundOffice>en
RouterOfLostandFoundOffice#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RouterOfLostandFoundOffice(config)#sh ip int br
^
% Invalid input detected at '^' marker.
RouterOfLostandFoundOffice(config)#ex
RouterOfLostandFoundOffice#
%SYS-5-CONFIG_I: Configured from console by console
RouterOfLostandFoundOffice#sh ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.1 YES manual up up
FastEthernet0/1 10.10.12.1 YES manual up up
Serial0/1/0 10.10.1.1 YES manual up up
Serial0/1/1 10.10.2.2 YES manual up up
Vlan1 unassigned YES unset administratively down down
RouterOfLostandFoundOffice#
RouterOfLostandFoundOffice#
RouterOfLostandFoundOffice#
RouterOfLostandFoundOffice#
RouterOfLostandFoundOffice#
RouterOfLostandFoundOffice#
Copy
Paste
Top


Router Of Information Office
Document was last saved: Just now
Physical
Config
CLI
Attributes
IOS Command Line Interface
RouterOfInformationOffice(config)#
RouterOfInformationOffice(config)#int fa0/0
RouterOfInformationOffice(config-if)#ip add
RouterOfInformationOffice(config-if)#ip address 10.10.20.1 255.255.255.0
RouterOfInformationOffice(config-if)#no sh
RouterOfInformationOffice(config-if)#no shutdown
RouterOfInformationOffice(config-if)#ip dhcp pool vo
RouterOfInformationOffice(config-if)#ip dhcp pool voice
RouterOfInformationOffice(config-if)#ip dhcp pool vo
RouterOfInformationOffice(config-if)#ip dhcp pool v
RouterOfInformationOffice(config-if)#ip dhcp pool voice ip
^
% Invalid input detected at '^' marker.
RouterOfInformationOffice(config-if)#ip dhcp pool voip
RouterOfInformationOffice(dhcp-config)#net
RouterOfInformationOffice(dhcp-config)#network 10.10.20.0 255.255.255.0
RouterOfInformationOffice(dhcp-config)#def
RouterOfInformationOffice(dhcp-config)#default-router 10.10.20.1
RouterOfInformationOffice(dhcp-config)#option 150 ip 10.10.20.1
RouterOfInformationOffice(dhcp-config)#ex
RouterOfInformationOffice(config)#tel
RouterOfInformationOffice(config)#telephony-service
RouterOfInformationOffice(config-telephony)#max-e
RouterOfInformationOffice(config-telephony)#max-ephones 30
RouterOfInformationOffice(config-telephony)#max-dn 30
RouterOfInformationOffice(config-telephony)#ip so
RouterOfInformationOffice(config-telephony)#ip source-address 10.10.20.1 port 2000
RouterOfInformationOffice(config-telephony)# au
RouterOfInformationOffice(config-telephony)# auto as
RouterOfInformationOffice(config-telephony)# auto assign 1 to 30
RouterOfInformationOffice(config-telephony)#ex
RouterOfInformationOffice(config)#
RouterOfInformationOffice(config)#tel
RouterOfInformationOffice(config-telephony)#ex
RouterOfInformationOffice(config)#tel
RouterOfInformationOffice(config)#telephony-service
RouterOfInformationOffice(config-telephony)#ephon
RouterOfInformationOffice(config-telephony)#ephone-dn 1
RouterOfInformationOffice(config-ephone-dn)#nu
RouterOfInformationOffice(config-ephone-dn)#number 5752400
RouterOfInformationOffice(config-ephone-dn)#ephone-dn 2
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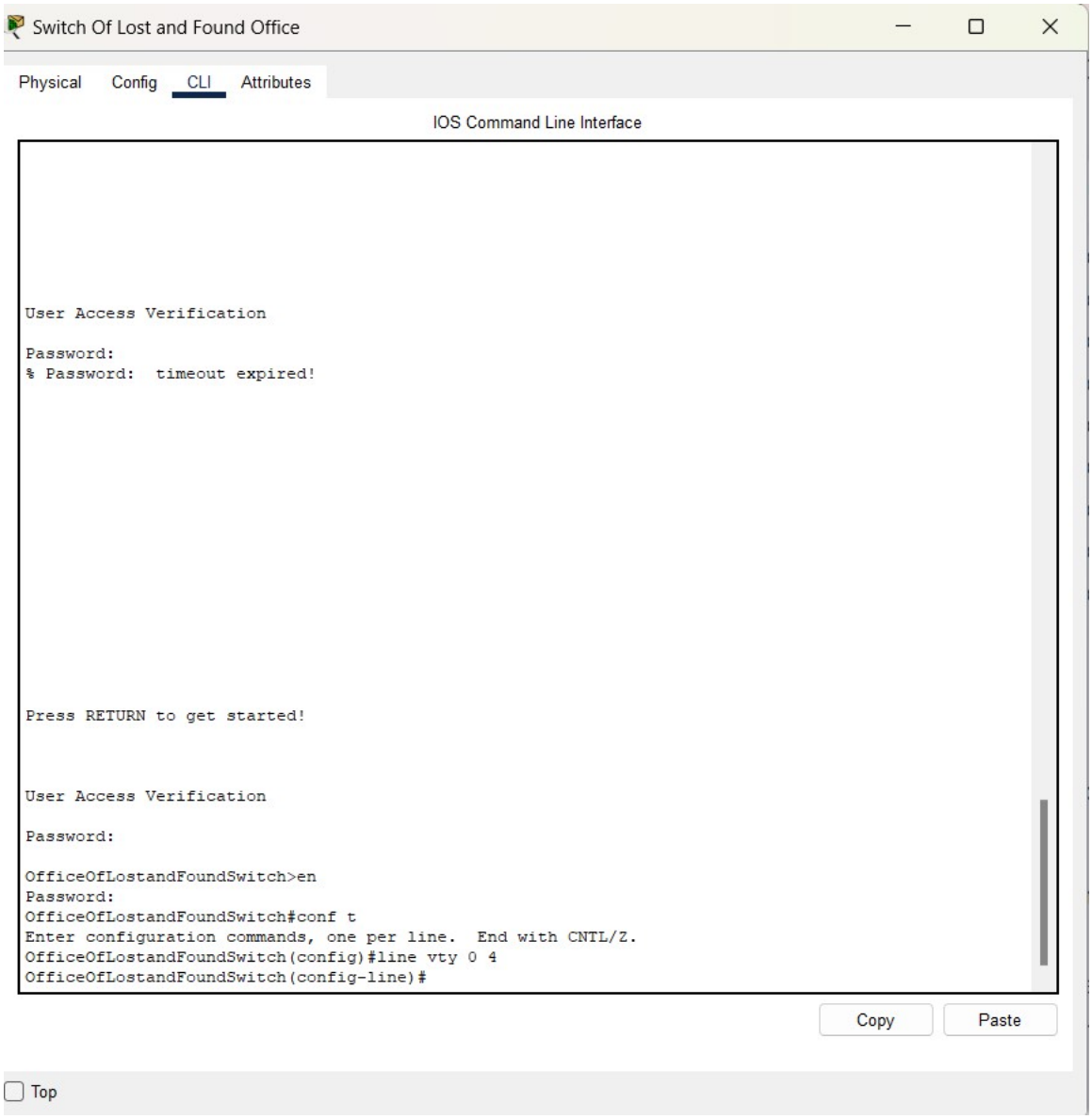
User password on Switch:

We set switchTicket for the user password, and set enterSwitch for the console password, then we encrypted both of them so when someone write show running config command, he cannot know the passwords.

- **In lostandfoundOffice: (router)**

```
RouterOfLostandFoundOffice>
RouterOfLostandFoundOffice>en
RouterOfLostandFoundOffice#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RouterOfLostandFoundOffice(config)#enable secret FFFF5-fghi6A
RouterOfLostandFoundOffice(config)#service password-encryption
RouterOfLostandFoundOffice(config)#interface gigabitEthernet 0/0
%Invalid interface type and number
RouterOfLostandFoundOffice(config)#do wr
```

(switch and console password):

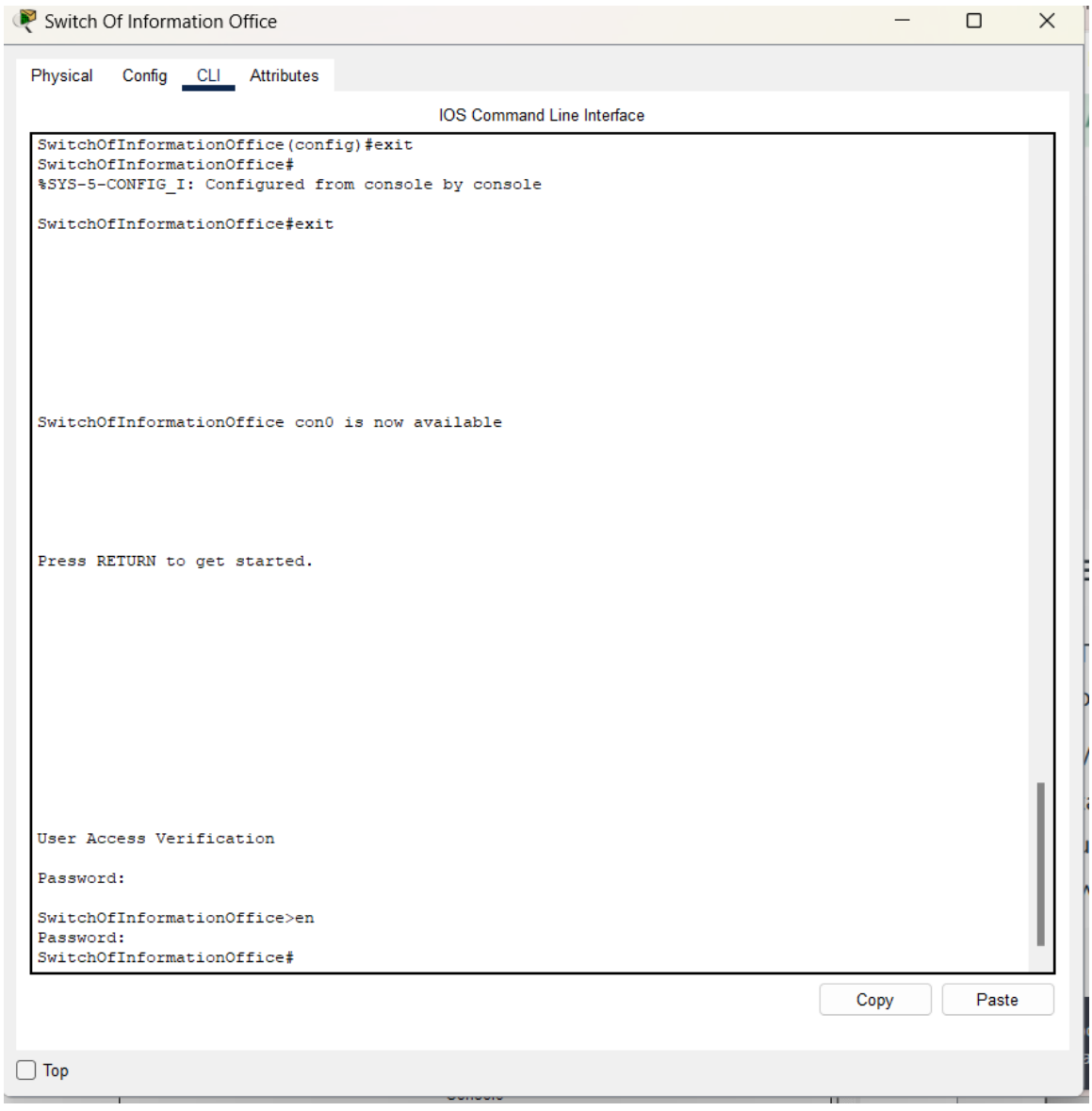


Consoul line :

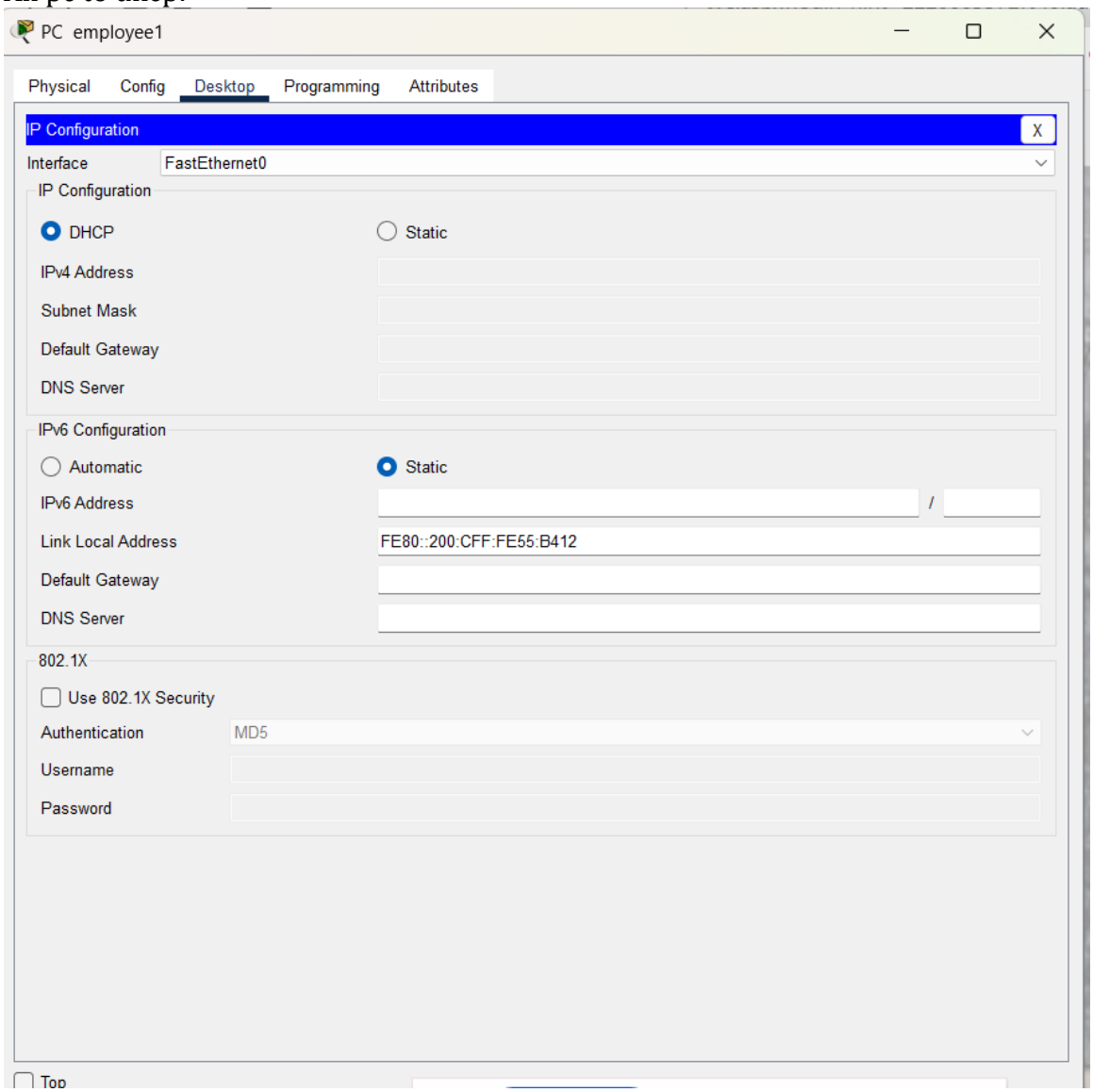
```
OfficeOfLostandFoundSwitch>en
Password:
OfficeOfLostandFoundSwitch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
OfficeOfLostandFoundSwitch(config)#line console 0
OfficeOfLostandFoundSwitch(config-line)#password FFFF4@fgh69-11
OfficeOfLostandFoundSwitch(config-line)#login
OfficeOfLostandFoundSwitch(config-line)#logging synchronous
OfficeOfLostandFoundSwitch(config-line)#exec-timeout 15 0
OfficeOfLostandFoundSwitch(config-line)#exit
OfficeOfLostandFoundSwitch(config)#line vty 0 4
OfficeOfLostandFoundSwitch(config-line)#password FFFF4@fgh69-11
OfficeOfLostandFoundSwitch(config-line)#login
```

```
OfficeOfLostandFoundSwitch(config-line)#logging synchronous
OfficeOfLostandFoundSwitch(config-line)#exec-timeout 15 0
OfficeOfLostandFoundSwitch(config-line)#exit
OfficeOfLostandFoundSwitch(config)#do wr
Building configuration...
[OK]
OfficeOfLostandFoundSwitch(config)#
OfficeOfLostandFoundSwitch(config)#exit
OfficeOfLostandFoundSwitch#
        %SYS-5-CONFIG_I: Configured from console by console
```

Another office :



All pc to dhcp:



Why dhcp ?
Because we are getting ip address dynamically

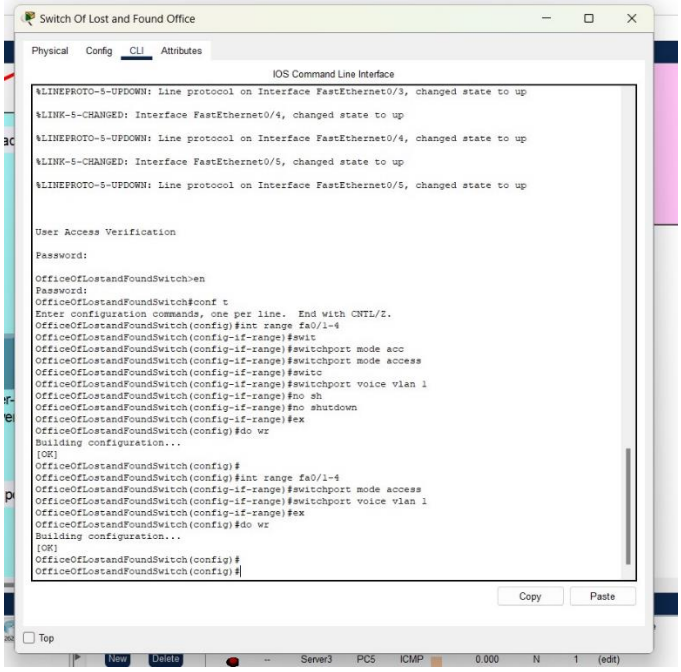
Password table for communication device:

Office	Interface	Password	Console line from :	Console password for switch
Office of lost and found	Privilege mode	Router → FFFF5-fghi6A sw p1→FFFF4@fgh69-12	set a password on this line from switch to router	FFFF4@fgh69-12
Information Office 1	Privilege mode	Router→ FFFF4-fgh69 Sw → FFFF4@fgh69	set a password on this line from switch to router	FFFF4@fgh69-11
Information Office 2	Privilege mode	Router → FFF5-fghi6A sw →FFFF5@fghi6A	set a password on this line from switch to router	FFFF4@fgh69-11

Vlan on Switches:

Here we activate the vlan 1 on all switches in the all office to facilitate and success sending messages from and to this office.

Information office :

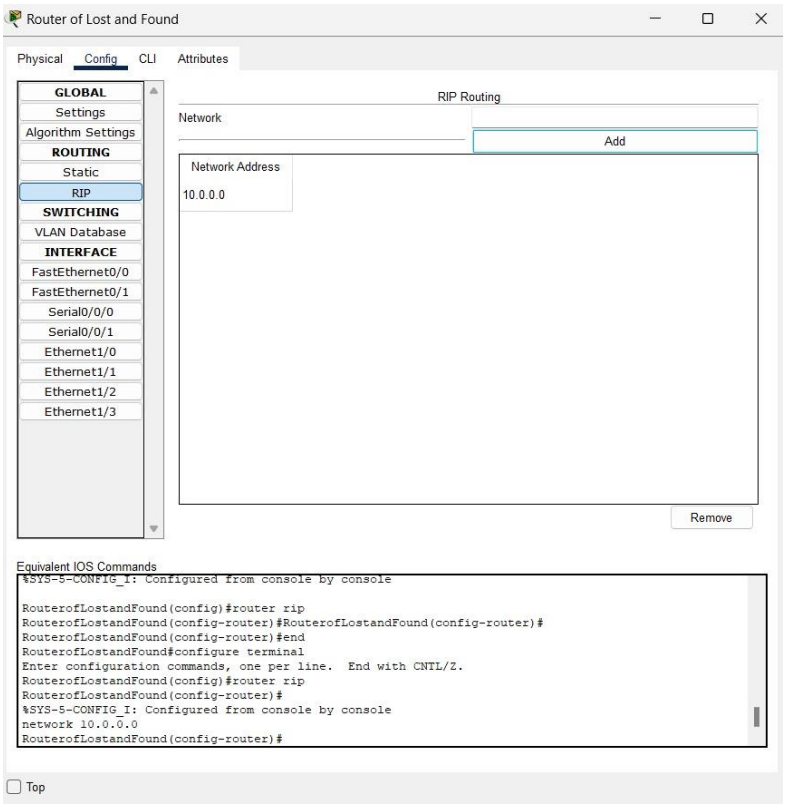


- RIP configuration.

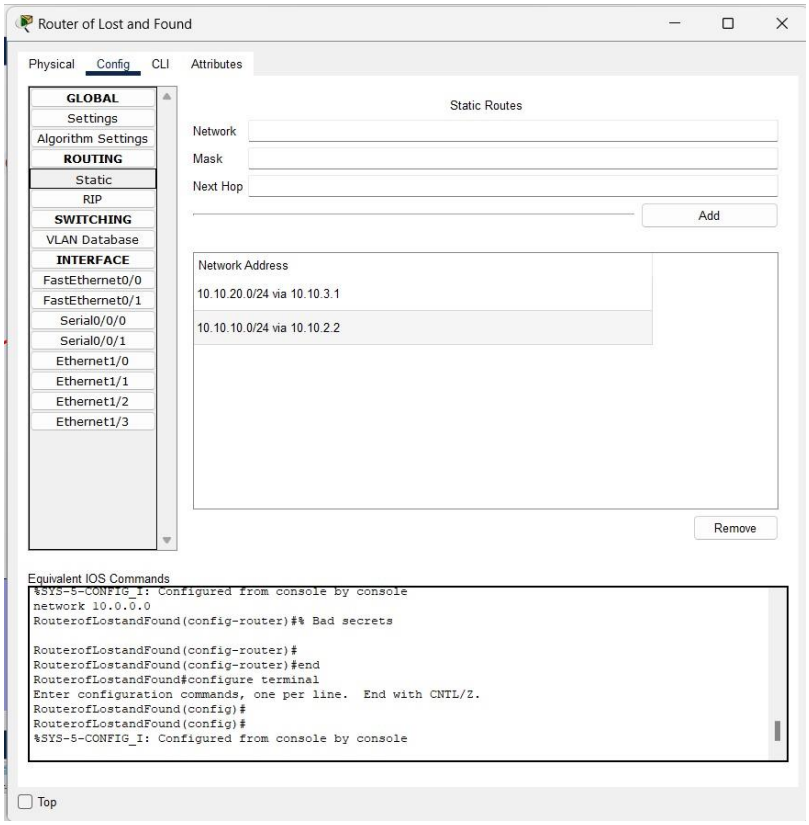
We have used this protocol in all our offices. We used both Static and RIP:

As here in router of Lost and Found office:

RIP:

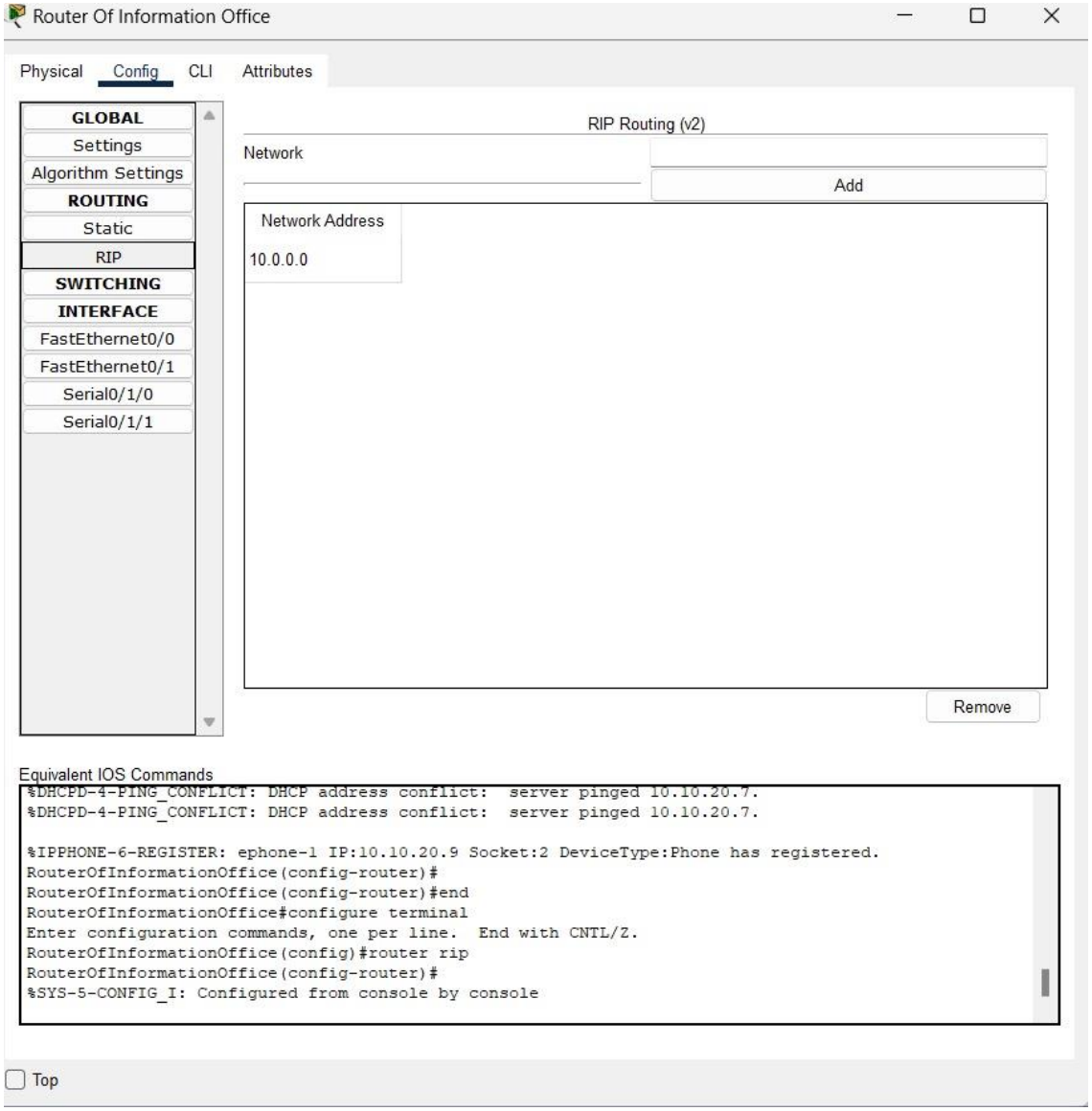

Router of Lost and Found
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Ethernet1/0
Ethernet1/1
Ethernet1/2
Ethernet1/3
RIP Routing
Network
Add
Network Address
10.0.0.0
Remove
Equivalent IOS Commands
%SYS-5-CONFIG_I: Configured from console by console
RouterofLostandFound(config)#router rip
RouterofLostandFound(config-router)#RouterofLostandFound(config-router)#
RouterofLostandFound(config-router)#end
RouterofLostandFound#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterofLostandFound(config)#router rip
RouterofLostandFound(config-router)#
%SYS-5-CONFIG_I: Configured from console by console
network 10.0.0.0
RouterofLostandFound(config-router)#
Top

Static:

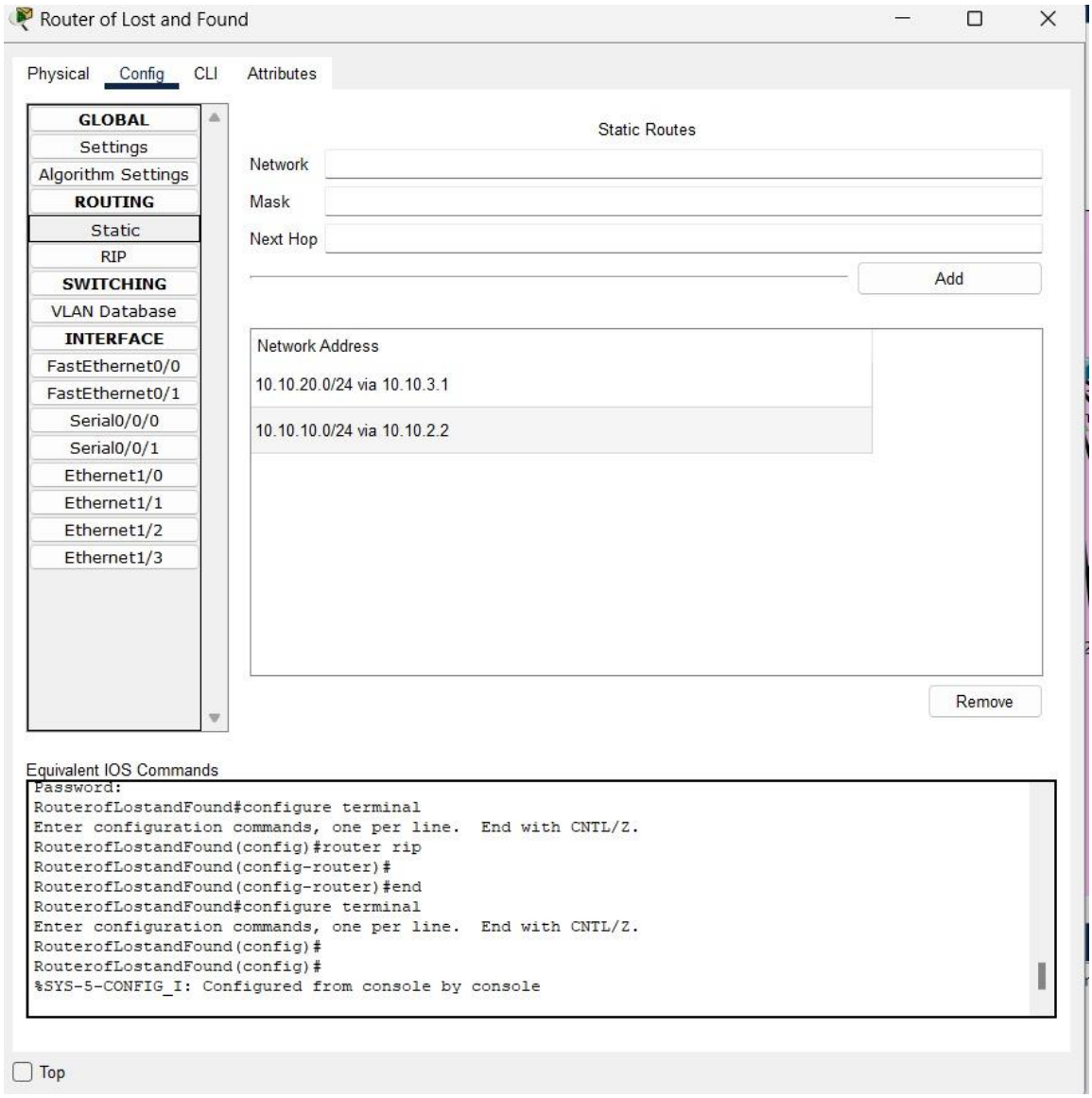

Router of Lost and Found
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Ethernet1/0
Ethernet1/1
Ethernet1/2
Ethernet1/3
Static Routes
Network
Mask
Next Hop
Add
Network Address
10.10.20.0/24 via 10.10.3.1
10.10.10.0/24 via 10.10.2.2
Remove
Equivalent IOS Commands
%SYS-5-CONFIG_I: Configured from console by console
network 10.0.0.0
RouterofLostandFound(config-router)#% Bad secrets
RouterofLostandFound(config-router)#
RouterofLostandFound(config-router)#end
RouterofLostandFound#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RouterofLostandFound(config)#
RouterofLostandFound(config)#
%SYS-5-CONFIG_I: Configured from console by console
Top

And in Information office:

RIP:



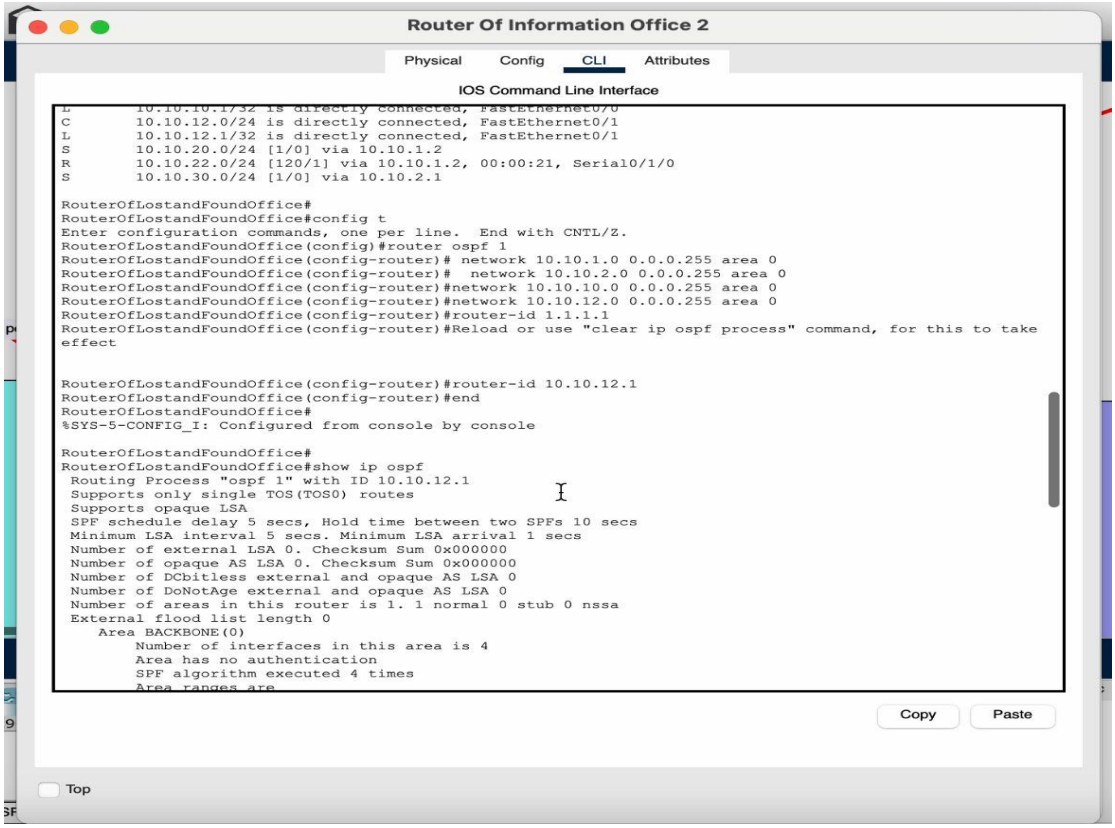
Static:



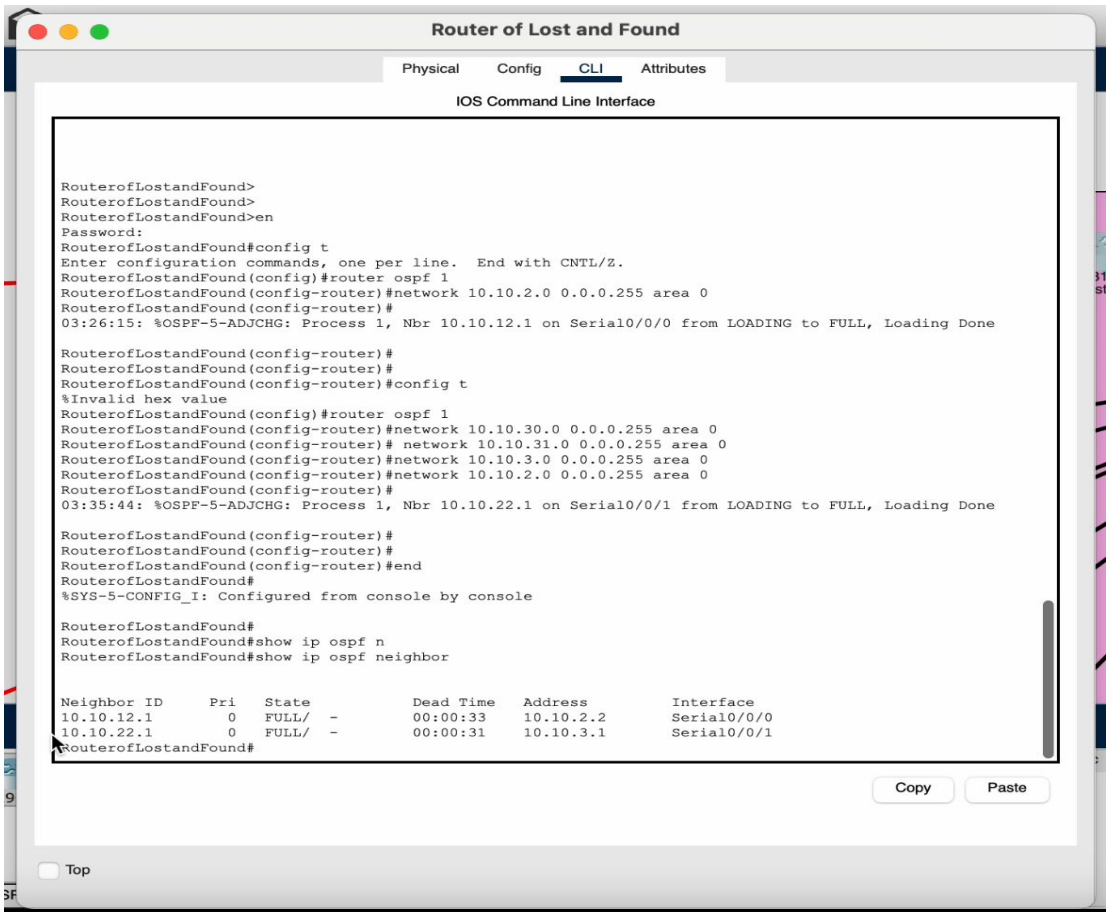
we did the same in the other information office.

- OSPF:

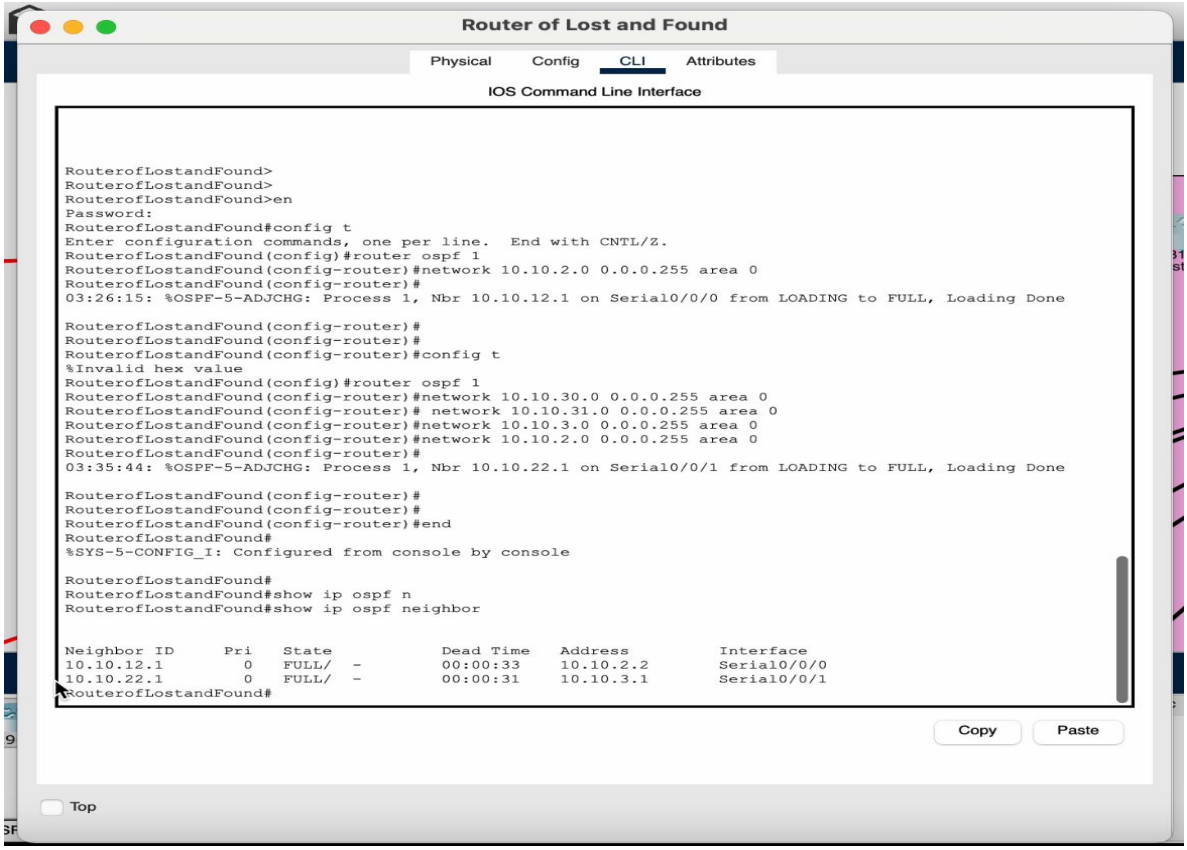
  Router of information office2:



Router of information office:



Router of lost and found



Compare between : RIP OSPF

	RIP	OSPF
Packet loss	0	0
CPU utilization	2.3%	34%
Memory usage.	38 MB	18.5 MB
Execution time	3.5 seconds	2 seconds

In our study, we compared the RIP and OSPF routing protocols for our crowded ring topology network. After analyzing their performance, we chose OSPF due to its superior suitability for the network's crowded transmission. OSPF's efficient routing and support for complex topologies make it the preferred choice, enhancing crowd management during Hajj and Umrah.

Conclusion :

In summation, our initiative encompasses the implementation of state-of-the-art sensor-based network infrastructure, incorporating advanced protocols to facilitate seamless connectivity among relevant offices and integrated transportation services management. This infrastructure enables real-time data collection on crowd dynamics, facilitating the identification of congestion hotspots and the implementation of proactive crowd management strategies. Moreover, it fosters efficient communication and coordination among various stakeholders involved in Hajj logistics, thereby enhancing operational efficiency and responsiveness. By leveraging advanced routing algorithms and real-time updates, we aim to optimize transportation routes and schedules, streamlining pilgrim movement within the holy sites. Ultimately, our comprehensive approach is designed to enhance the pilgrim experience, optimize crowd management .

References:

- Optimizing Pilgrim Flow: Sensor Integration and Real-time Analysis for Umrah and Hajj Management"-network report
- -Felemban, E. A., Rehman, F. U., Biabani, S. A. A., Ahmad, A., Naseer, A., Majid, A. R. M. A., ... & Zanjir, F. (2020). Digital revolution for Hajj crowd management: a technology survey. IEEE Access, 8, 208583-208609

- -Rahman, M. A., & Kabir, A. I. (2023). A network design and simulation of a Smart Railway Station using Cisco Packet Tracer.

- Santana, J. R., Sánchez, L., Sotres, P., Lanza, J., Llorente, T., & Muñoz, L. (2020). A privacy-aware crowd management system for smart cities and smart buildings. IEEE Access, 8, 135394-135405.

- Wang, A., Guo, Y., Hao, F., Lakshman, T. V., & Chen, S. (2014, December). Scotch: Elastically scaling up sdn control-plane using vswitch based overlay. In Proceedings of the 10th ACM International on Conference on emerging Networking Experiments and Technologies (pp. 403-414).

- giriteja94495.(May 1, 2021) . Git statsGarage-Control-Simulation-Using CISCO_PACKET_TRACER . github.com.